

Nathaniel K. Green

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Education

George Mason University

Spring 2026

Graduate non-degree seeking

GPA: 4.0 (out of 4.0)

Coursework: Topology of metric spaces and manifolds

University of Wisconsin–La Crosse

May 2025

B.S. in Mathematics (Honors) and Computer Science

GPA: 3.83 (out of 4.0)

Mathematics coursework: Abstract Algebra, Real Analysis, Topology, Galois Theory, Functional Analysis, Complex Variables, ODEs, Linear Algebra, Calculus I, II, and III

Computer Science coursework: Theory of Computation, Machine Learning, Artificial Intelligence, Introduction to Compilers

Research Interests

Functional analysis, operator theory, linear algebra, topology, algebraic geometry

Publications

- **Topological Structure of the space of composition operators on the Lipschitz space of a tree.**
With Robert F. Allen. *In Progress*, 2026.
- **Corrigendum: [Topological structure of the space of composition operators on L^∞ of an unbounded, locally finite metric space].**
With Robert F. Allen. *In Progress*, 2026.
- **Further techniques on a polynomial positivity question of Collins, Dykema, and Torres-Ayala.**
With Edward D. Kim. *Algebra and Discrete Mathematics*, 2024. PDF

Talks and Posters

- **Poster:** *CDTA Conjecture* – UW-La Crosse Research & Creativity Symposium (2023)
- **Poster:** *Positive Results on an Open Case of a Conjecture Concerning Polynomials and Matrix Operations* – UW-La Crosse Summer Research Symposium (2022)
- **Talk:** *Dean’s Distinguished Fellowship Research Presentation* – UW-La Crosse (June 2022)

Honors and Awards

- Undergraduate Research & Creativity Laureate, UW-La Crosse (2025)
- Honors Degree in Mathematics, UW-La Crosse (2025)
- Undergraduate Research Scholar (2025)
- Undergraduate Research Committee Grant (Spring 2023)
- Dean’s Distinguished Fellowship (2022)
- Dean’s List, multiple semesters (2021–2025)

Academic Service

- Treasurer, Math and Stats Club, UW-La Crosse
- Volunteer peer tutor (informal), UW-La Crosse

Industry Experience

Capital One — Software Engineer Aug 2025 – Present

- Working in fraud detection for Wire and ACH transfers.
- Implemented Claude and Windsurf workflows to automate general tasks and increase productivity.

Capital One — Software Engineer Intern June 2024 – Aug 2024

- Built an automated metadata migration pipeline using AWS Lambda, S3, SNS, and CloudWatch.

Technical Skills

- Languages: Python, Java, C, JavaScript, SQL, HTML/CSS, LaTeX
- Tools: Git, AWS, Claude, Windsurf, VSCode, Linux/Unix, Jira

References

Available upon request